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Simulation of molten pool dynamics and stability analysis in laser buttonhole welding

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Abstract

Molten pool dynamics in laser buttonhole welding with wire feeding and laser beam oscillation were studied using computational fluid dynamics (CFD). The simulation results, such as temperature fields and molten pool velocities, were used to analyze the characteristics of molten pool dynamics. The dynamics in the rear part of the molten pool area were significantly reduced in comparison to those in the front when a buttonhole (a cavity in the molten pool) was present.

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Keywords: numerical simulation; laser welding; beam oscillation; buttonhole welding; seam quality

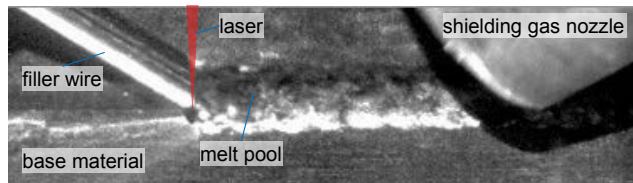
1. Introduction

For the autogenous laser welding of butt joints, the alignment of the sheets should be precise enough to enable gap bridging by the seam and, as result, a high weld quality and joint strength. The usage of additional filler wire can significantly increase the gap bridging ability, and the filler material can purposefully affect the metallurgical seam properties. However, misalignments of the wire can still cause seam defects. A laser welding process with wire feeding in combination with beam oscillation has been suggested to compensate for typical positioning errors of the wire [1]. Without the use of a filler wire, it has also been reported that weld seam defects could be reduced by using an oscillating laser beam [2]. In laser welding of stainless steel sheets without beam oscillation, a big hole was observed [3], whereby the process was named laser donut welding [3]. The catenoidal shaped hole helps to improve the weld quality in pulsed welding, but in continuous wave welding it is recommended to avoid its formation as it could result in discontinuities in the solidified seam [4]. Seam discontinuities were also observed in laser welding of aluminum alloy sheets

[5]. A similar phenomenon – the so-called buttonhole – was observed in laser welding of aluminum sheets with an air gap with the use of wire feeding and transversal beam oscillation [6]. The gap bridging ability was significantly increased by the process design [7]. With a stable buttonhole formation, straight courses and an almost complete lack of ripples were observed in solidified weld seams [8]. At an oscillation frequency of 100 Hz, a keyhole presented, but it did not grow to the size of the buttonhole with its catenoidal shape; see Fig. 1 [9]. The stable buttonhole was observed at the frequency of 250 Hz; see Fig. 1 [9]. The formation of the buttonhole could be simulated by computational fluid dynamics (CFD) with realistic solid wire feeding and beam oscillation [9], whereby the shape of the wire tip seems to be important for the stability of the buttonhole [9]. The length of the wire tip (L_t) was shorter and the wire melting angle (Θ) between the wire front and the welding direction was larger at the frequency of 250 Hz; see Fig. 2 [9]. The calculated fusion zone profiles are in good agreement with the experimental results [9].

The reason for the better surface quality in the case of a stable buttonhole at the oscillation frequency of 250 Hz has

not yet been fully understand. Thus, the molten pool dynamics in buttonhole welding is investigated in this study by using CFD simulations.



a) Oscillation frequency 100 Hz



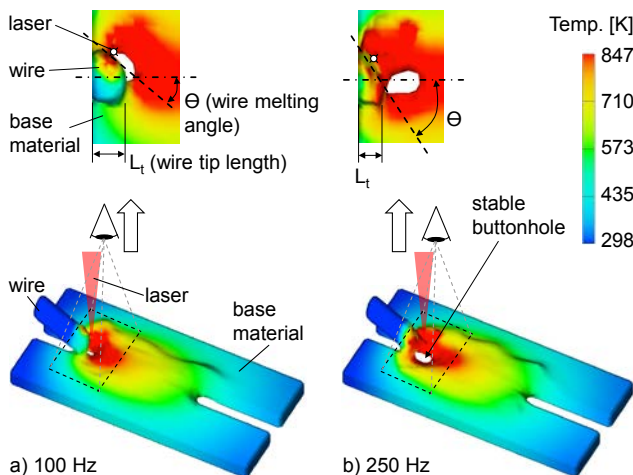
b) 250 Hz

laser power: 3 kW; welding speed: 6 m/min;
wire feed speed: 9 m/min; oscillating width 1.4 mm

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Fig. 1. High speed camera images of experiments [9]



a) 100 Hz

b) 250 Hz

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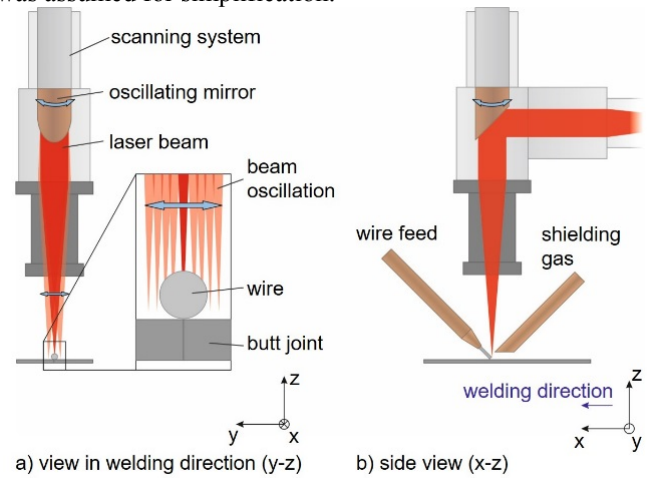
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Fig. 2. Simulation results in a three-dimensional view [9]

2. Method

An IPG fiber laser was used to weld aluminum alloy sheets (AlSi1MgMn) of 1 mm thickness with an air gap of 1 mm in butt joint configuration. Aluminum alloy filler wire (AlSi5) with 1.2 mm diameter was used to fill the gap. A laser beam focused on the upper surface of the sheets was oscillated transversal to the welding direction. In combination with the process feed, a sinusoidal-shaped path resulted on the workpiece. A schematic sketch of the process is shown in Fig. 3 and the welding process parameters are given in Table 1. Fig. 4 shows a schematic description of the computational domain with solid wire feeding and laser beam oscillation. The minimum mesh size used in the simulations was 0.1 mm. The total number of cells was 0.7 million. The thermo-physical material properties of pure aluminum and the aluminum alloy were used in numerical simulations, as described in a previous work [9]. In the computational domain, the governing equations of mass, momentum, and

energy conservation were solved to simulate the molten metal flows. The volume-of-fluid (VOF) method was adopted to calculate the temporal free surface shape. Finally, a single-phase incompressible laminar flow with Newtonian viscosity was assumed for simplification.



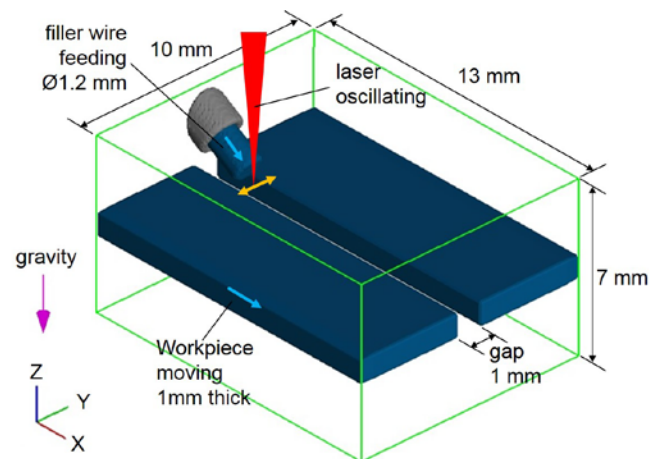
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Fig. 3. Schematic illustration of the experimental setup [9]

Table 1. Welding process parameters

Parameter	Values
Laser power	3.0 kW
Focus diameter	100 μ m
Focal position to sheet surface	0 mm
Welding speed	6.0 m/min
Wire feeding speed	9.0 m/min
Oscillation frequency	250 Hz
Oscillation width	1.4 mm



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Fig. 4. Computational domain of the laser welding process with wire feeding and beam oscillation [9]

A detailed description of the laser beam welding model can be found in [10]. In this study, 697 energy bundles with Gaussian-like distributions were predefined along the axis of the beam for ray-tracing. The ray-tracing algorithm was used to estimate the multiple reflections on the free surface. A Fresnel reflection model was used to calculate the reflectivity

of the circularly polarized laser beam on the material. Here, a simplified model using the Hagen–Rubens relation was not used due to the large discrepancy between the general and simplified models for fiber laser welding of aluminum. The recoil pressure, the buoyancy force, and the adiabatic bubble were considered, while the metal vapor and plasma dynamics were not considered.

Energy and pressure conditions were considered as boundary conditions on the free surface. The energy balance of the laser heat flux as well as the heat dissipation by convection and radiation are expressed in Eq. (1).

$$\begin{aligned} k \frac{\partial T}{\partial \vec{n}} &= q_L - q_{conv} - q_{rad} \\ &= q_L - h(T - T_0) - \varepsilon_r \sigma (T^4 - T_0^4) \end{aligned} \quad (1)$$

where, $\vec{n}$ is the vector normal to the local free surface, h is the convective heat transfer coefficient, T_0 is the ambient temperature, ε_r is the surface radiation emissivity, and σ is the Stefan-Boltzmann constant. The pressure balance is expressed as shown in Eq. (2).

$$p = p_r + \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (2)$$

where, p_r is the recoil pressure, γ is the surface tension, and R_1 and R_2 are the principle radii of the surface curvature by the Young-Laplace equation. If the surface tension pressure is zero ($R_1 = -R_2$), a catenoid can theoretically be formed. This has also been observed in experiments [3]. The catenoidal-shaped hole is larger than a typical keyhole and can be maintained without the recoil pressure. All the governing equations were solved by finite-difference and finite-volume approximations using the Flow3D CFD commercial software [11]. Simulating 0.1 s of laser welding requires a wall clock time of about 33 hours with an Intel® Xeon® CPU E5649 and 48 GB RAM.

3. Results

Fig. 5 and Fig. 6 show the cross-sectional side and top views of the temperature and z-velocity distributions at the welding process time of about 0.03 s in sequence over time.

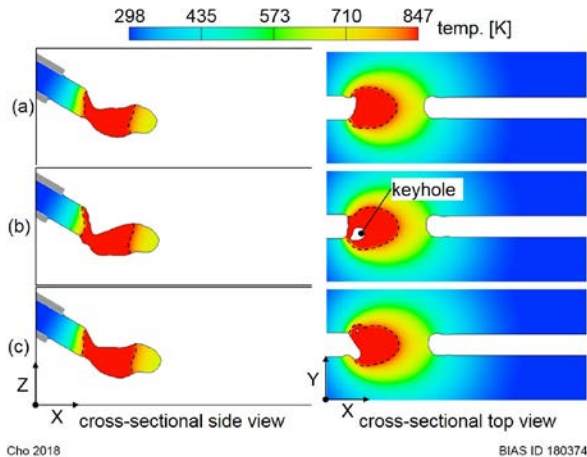


Fig. 5. Cross-sectional temperature views over time at the beginning of the process without buttonhole (a): 0.032 s; (b): 0.033 s; (c): 0.034 s

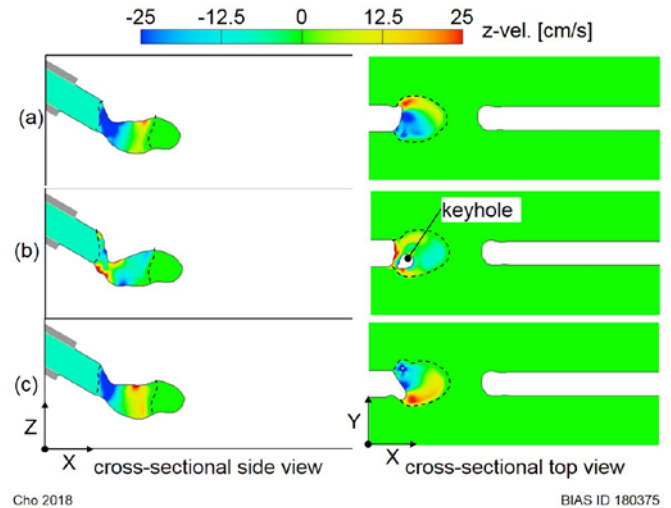


Fig. 6. Cross-sectional z-velocity views over time at the beginning of the process without buttonhole (a): 0.032 s; (b): 0.033 s; (c): 0.034 s

There was no buttonhole formation at the beginning of the process. The molten pool near the keyhole fluctuates severely in the z-direction and shows high z-velocities. The rear part of the molten pool also has high dynamics.

Fig. 7 and Fig. 8 show the cross-sectional side and top views of the temperature and z-velocity distributions at the welding process time of about 0.09 s in sequence over time. The developed buttonhole can be seen in the simulation result. Based on the position of the buttonhole, the molten pool is divided into a front and a rear part. The front molten pool fluctuates severely in the z-direction and shows high z-velocities. In comparison with the front, the rear part of the molten pool is calmer, as demonstrated by the relatively smaller velocities. The buttonhole always has a circular shape in the rear part but is partially distorted in the front part.

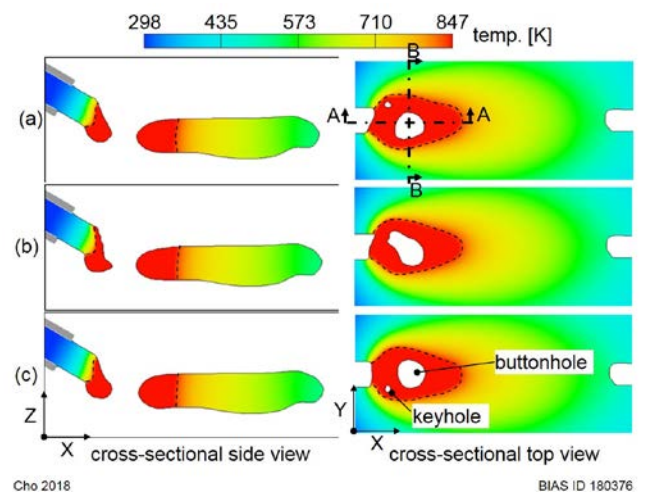


Fig. 7. Cross-sectional temperature views over time (a): 0.094 s; (b): 0.095 s; (c): 0.096 s (the so-called front part of the molten pool is on the left and the so-called rear part is on the right of the buttonhole in the views; section planes A-A and B-B are given in Fig. 9)

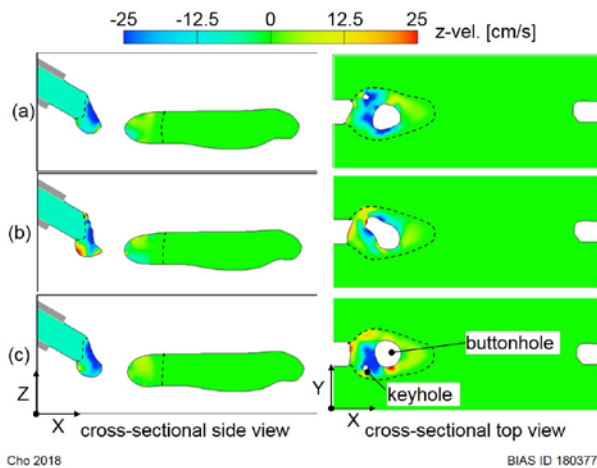


Fig. 8. Cross-sectional z-velocity views over time (a): 0.094 s; (b): 0.095 s; (c): 0.096 s (the so-called front part of the molten pool is on the left and the so-called rear part is on the right of the buttonhole in the views)

Fig. 9 shows the cross-sections of the buttonhole presented in Fig. 7a. The buttonhole geometry is similar in the longitudinal cross-section A-A and in the transverse cross-section B-B. The dashed lines in Fig. 9 depict a theoretical catenoidal shape according to [3]. The calculated cross-sections have wider uppermost and lowermost openings than the theoretical catenoid with the same radius in the center of the catenoidal hole. The section A-A is inclined at 10 degrees in comparison with the normal theoretical shape.

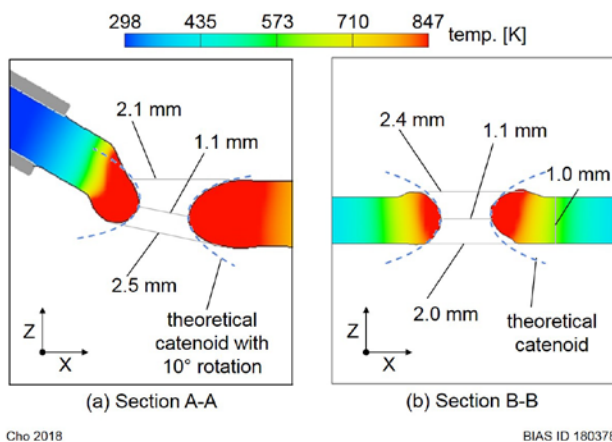


Fig. 9. Cross-sections of the simulated buttonhole in Fig. 7a in comparison with a theoretical catenoidal shape according to [3]

4. Discussion

Without the existing buttonhole, the high dynamics of the oscillating keyhole were transferred directly to the rear part of the molten pool. Thus, the dynamics of the laser deep penetration welding process with beam oscillation affect the surface quality of the solidified seam.

In the case of a stable buttonhole, the molten pool calms down significantly in the rear part. The buttonhole shape was distorted at the front side due to the keyhole. The circular shape was maintained at the rear part. It appears that the buttonhole acts like a vibration damper by dividing the molten pool into a highly dynamic front part and a calm rear part. The stable buttonhole shape was similar to the so-called donut

hole shape according to [3], which could maintain itself even after the laser had been turned off. The buttonhole had wider openings than a theoretical catenoid, which means it could be stable. It seems that the buttonhole is also a self-sustaining hole with a catenoidal shape, and thus it can partially resist disturbances, e.g., high dynamics from the front part of the molten pool.

5. Conclusion

The effect of the buttonhole on the molten pool dynamics can be shown with CFD simulations. The high dynamics in the front molten pool induced by the oscillated laser beam with keyhole formation are significantly calmed down in the rear part by the buttonhole. Thus, the damping effect of the catenoidal-shaped hole in the molten pool is probably the reason for the high surface quality achieved in buttonhole welding.

Acknowledgements

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